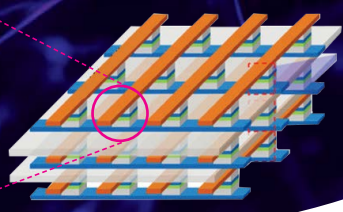
MIMICKING THE BRAIN

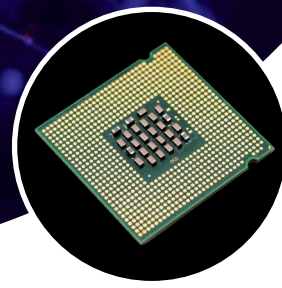
Still in nascent stages, neuromorphic computing will complement traditional computers to reimagine the way machines work



2 IISc researchers have developed a computing accelerator, comprising a grid of memory storage devices called memristors, which mimic this biological synaptic function



3 This allows for faster processing by storing data in 16,500 distinct states. This is unlike a traditional computer chip which has only two conductance states as a digital code is broken down to a binary of zeroes and ones



4 Using the accelerator, the IISc team recreated NASA's iconic image of 'Pillars of Creation' from the raw data of the James Webb Space Telescope in 2022. It rendered 1,664,000 pixels in 26,000 steps. A traditional computer chip would require 100 million steps



5 By drastically cutting down computational steps, the accelerator allows complex tasks to be done locally. Besides speeding up computing, this will save energy and ensure data security

of zeroes and ones. What this essentially means is a vast number of calculations can be reduced to only a few steps, perhaps even a single one.

Revolution in the Making

In the late 1990s, the graphics processing unit (GPU) was the technology breakthrough that fired up the world of image processing and Artificial Intelligence (AI) through parallel processing—a computing method that uses a large number of cores within the chip, each doing a part of the overall task. The IISc accelerator, says IT industry veteran Kris Gopalakrishnan, can potentially "disrupt" the GPU ecosystem. "Using analogue circuitry created through materials that [Goswami et al] have researched on, they have created this memristor which allows you to do matrix multiplication in one step, which normally in the traditional computer takes multiple steps," he explains. Memristors are the key component of the accelerator. A resistance switching memory device, it stores and computes data at the same location unlike digital computer chips which move data back and forth between a memory unit and a processing unit. "So that would reduce your computing energy consumption and time, manyfold," explains Goswami.

Both energy and time are crucial in today's compute-intensive era of AI where machines are trained on huge volumes of data. Typically, this involves transporting all this data to a data centre, or the 'cloud'. But with the memristor, many AI tasks can be done locally. "It's AI on the Edge. You can have large language models trained on the Edge and not in the data centres," says Prof. Navakanta Bhat, who led the circuit development and testing. Edge here refers to servers located midway between a device and a data centre, which thereby reduces the otherwise long

THE IISc BREAKTHROUGH PROMISES TO PLACE INDIA ON THE GLOBAL MAP OF COMPUTING INVENTIONS

Graphic by NILANJAN DAS & TANMOY CHAKRABORTY